

YugabyteDB: The Reasons Why It's Popular

Yugabyte was first developed as closed source commercial software.

However, by 2019, its founders realised the power of the open source community and made it available under the Apache 2.0 license. The timing was right as developers were on the lookout for a cloud native database. The adoption and popularity of this database has increased significantly in recent years for a number of reasons.



YugabyteDB was developed by former Facebook software engineers nearly six years ago. It is today widely adopted across enterprises and industries as it is a fully functional open source distributed SQL database, which supports cloud-native applications.

Distributed SQL acts as a single logical database, whereas behind the scenes it is a database cluster of one or more nodes. A distributed SQL database has a three-layer architecture:

- **SQL API** is the primary interface for applications to model relational data. Like RDBMS indexes, foreign key constraints, JOIN queries and multi-row ACID transactions are unique data models for distributed SQL.
- **Distributed query** execution takes the responsibility for distributed query processing by sharing query requests across other nodes in the cluster. The execution across multiple nodes minimises latency and keeps a check on the amount of data transferred amongst them.
- **Distributed data storage** ensures

data is distributed and replicated, avoiding bottlenecks and ensuring high availability, leading to high performance.

Figure 1 gives a reference implementation of distributed SQL across three nodes. An application connecting to node 1 leverages node 2

and 3 for query processing. In case node 1 goes down, node 2 or node 3 picks up the request and continues to process the application, ensuring high availability at all times.

There are many distributed SQL implementations. YugabyteDB is one of the first and most successful ones

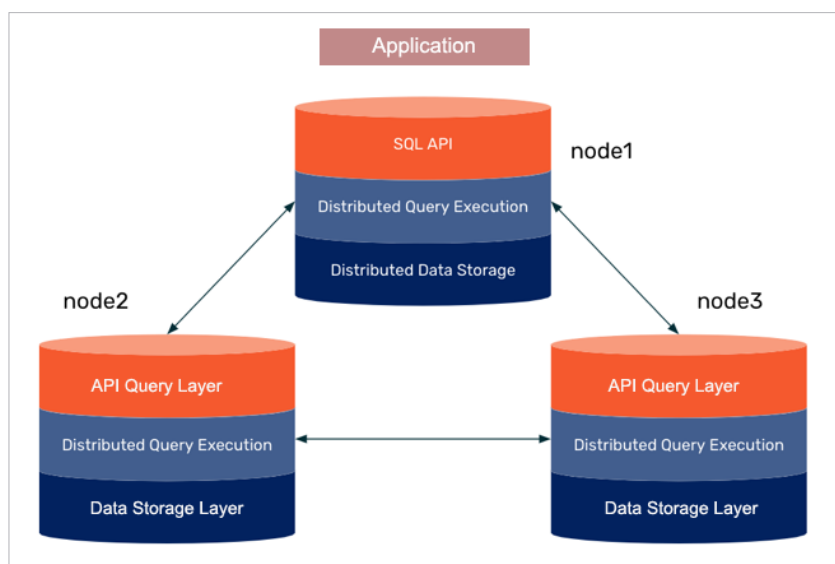


Figure 1: Distributed SQL architecture